Critical factors in the reconstruction of 3D models using the Roofer algorithm: an analysis of geometric and topological consistency in mountainous cities

1. Results

The results of the analysis, structured according to the critical factors identified throughout the reconstruction process, are presented. These factors can be broadly organized into two main categories: (i) those related to the quality and coherence of the input data (factors 1, 2, 3, 4, and 5), and (ii) those associated with the physical and constructive conditions of the urban environment (factors 6, 7, 8, and 9). This classification provides a systematic framework for understanding how different sources of variability influence the reconstruction outcomes.

Table 1 summarizes the identified critical factors, their effects on the process, and the proposed methodological adjustments to obtain geometrically consistent and topologically valid 3D models.

Workflow stage	**Code**	**Critical factor**	**Observed problem**	**Effect on 3D model**	**Methodological adjustment**
Stage 1 – Geometric control of footprints	CF1	Geometrical inconsistencies in footprints	Self-intersections, nested rings, spikes, collapsed geometry	Failures in reconstruction or generation invalid geometries	Preliminary validation and topological correction
	CF2	Footprints with 3D information	Footprints from restitution with inconsistent 3D vertices	Distortion at the base and surface of the building reconstruction	Remove Z component
	CF3	Inconsistent internal building subdivisions	Incorrect building delineation	Generation of multiple solids or inconsistent extrusions (table legs)	Input review and spatial cleaning
Stage 2 – Lidar altimetric preprocessing	CF4	Quality of lidar classification	Incorrect point classification in building class	Roof plane detection failures, missing planes, non-extruded constructions, and topological errors in the mode	Refine building class and adjust reconstruction parameters such as *clip-terrain*
	CF8	Polycarbonate roofs	Ground points beneath this type of construction	Reduced reconstruction quality	Remove ground points beneath buildings and adjust reconstruction parameters such as *clip-terrain*
	CF9	Local construction characteristics	architectural features not reconstructed	Loss of geometric fidelity	Evaluate the impact of the reconstruction based on the case study; for tanks, it is recommended to classify them as “building” and adjust the parameters for their reconstruction
Stage 3 – Alignment verification	CF5	Misalignment between footprints and lidar	Differences between data sources and footprint clipping errors	Displaced volumes and generation of geometric artifacts	Adjust geometry and location of footprints prior to reconstruction
Stage 4 – Parameter configuration	CF6	Presence of vegetation	Omission of buildings due to occlusion	Non-extruded constructions	Adjust reconstruction parameters (e.g., plane-*detect-min-points, clip-terrain, plane-detect-k.)* Or it is recommended to increase the density of lidar data collection
	CF7	Estimation of base elevation in slope areas	Buried constructions or constructions adapted to slope not represented	Incorrect base elevation Z_0	Compute Z_0 using a smaller buffer and define it as an input parameter *h-terrain-attribute*

Table 1. Summary of critical factors in 3D reconstruction

1.1 Critical factor 1: Geometric inconsistencies in footprints

Errors associated with the geometry of building footprints used as input data were identified, which prevented the execution of the algorithm. This behavior demonstrated that the geometric quality of footprints is a direct constraint on the feasibility of the 3D reconstruction process. The footprints used in this study were obtained from official sources of basic cartography and cadastral databases in Colombia, which have been previously validated according to the technical standards of the Agustín Codazzi Geographic Institute (IGAC). However, the results show that such validation does not guarantee their suitability for three-dimensional modeling processes, as geometric inconsistencies were identified that, although they do not affect their use in 2D environments, lead to failures in algorithms that require strictly valid geometries.

In particular, errors such as self-intersections and nested interior rings were detected, which violate the geometric validity rules defined by the OGC Simple Feature Access standard (OGC, 2021). These rules establish that polygons must be topologically closed, without intersections between rings, without degenerate geometries, and with a single connected interior. Failure to meet these conditions prevents footprints from being correctly interpreted as valid surfaces, directly affecting the 3D reconstruction process. To address this issue, a verification and correction process was carried out using geometric validation tools in QGIS (Check Validity and Fix Geometries), complemented by manual editing in specific cases (Kukulska et al., 2018).

1.2 Critical factor 2: Footprints with three-dimensional information

During the first iterations of the reconstruction process, it was identified that the presence of three-dimensional information in building footprints generated geometric inconsistencies. In particular, when using IGAC footprints that contain Z coordinates in their vertices, the appearance of invalid surfaces and geometric deformations in the reconstructed models was observed. These inconsistencies are not directly observed in the original footprints, but manifest during the three-dimensional reconstruction process.

As a result, geometric artifacts such as spikes, downward extrusions and deformations in roofs were observed, which generated incorrect reconstructions in some cases. The verification of the data made it possible to identify that some vertices of the footprints contained altimetric values equal to zero. In the context of the study area, with an approximate altitude of 2780 m above sea level, this situation acquires critical relevance, since these values introduce significant altimetric discrepancies during the reconstruction process. During the reconstruction process, it was evidenced that the altimetric values associated with the vertices of the footprints influenced the definition of the base and the roofs of the buildings, which resulted in geometric distortions in the resulting model. Additionally, inconsistencies associated with photogrammetric restitution practices used in footprint generation were identified, where the tracing of buildings is not vertex to vertex, but corresponds to simplified representations that do not adequately represent the three-dimensional geometry of the construction. As a consequence, footprint vertices contain elevation values that do not belong to the same plane, introducing small altimetric variations that hinder the generation of coplanar surfaces during the reconstruction process.

Additionally, a high frequency of type 203 errors associated with deviations from surface planarity was identified. In models generated from 3D footprints, these deviations reach values of up to 27 m, which determines a significant break in face coplanarity. In contrast, in 2D footprints only one error of this type was recorded with a deviation of 0.03 m, a value within the expected range of lidar noise or plane fitting.

1.3 Critical factor 3: Inconsistent internal buildings

Inconsistencies associated with the delimitation of buildings from an alternative data source, corresponding to the cadastral database, were also identified. When using this layer as input data, geometric errors were found in the reconstructed models, which manifested mainly as vertical extrusions, "*table legs*" type, generating inconsistent solids. The review of the data showed that the observed table legs were not associated with reconstruction parameters nor with errors in lidar data classification, but with inconsistencies in the internal delimitation of constructions in the cadastral layer. In several cases, the same building was represented through partial subdivisions and with internal polygons, which did not correspond to complete building footprints. Since these internal delimitations do not represent the real extent of the building nor contain ground points around them, the algorithm is not able to correctly identify the base surface associated with the terrain, which produces anomalous extrusions in the model geometry. The absence of topological connectivity with the terrain in internal footprints induces the algorithm to extrapolate the volume to lower levels, generating table legs.

1.4 Critical factor 4: Quality of lidar classification

During the reconstruction process, it was identified that incorrect classification of points in the lidar point cloud, particularly in the building class, directly affects the ability of the algorithm to adequately detect roof planes. This problem manifests mainly in the intersection zones between roof planes, which represent key areas for the Roofer algorithm during the surface detection and fitting process. As a consequence, failures occur in plane detection, resulting in partial constructions, non-extruded elements and the presence of topological errors in the model.

In the performed tests, it was observed that several of the errors identified by val3dity, particularly errors 102 (consecutive identical points) and 303 (non-manifold), were significantly reduced after refining the lidar classification in the building class. This adjustment allowed improving roof plane detection and reducing geometric inconsistencies in the reconstructed models. On the other hand, the presence of error 302 (non-closed shell) was identified, mainly associated with holes located at building edges. This behavior is related to the clipping procedure performed by the algorithm when it detects points classified as ground. In these cases, the presence of misclassified points induces improper clipping, generating holes in the reconstructed models.

To evaluate possible solutions, tests were carried out adjusting the *clip-terrain* parameter of the Roofer algorithm to false, in order to avoid automatic clipping of the roof when points classified as ground appear within or near it. This adjustment proved especially useful in point clouds such as the one used in this study, where the initial classification did not present optimal quality. The combination of refining lidar classification and adjusting the *clip-terrain* parameter made it possible to improve the geometric regularity of reconstructed buildings and reduce the propagation of errors derived from incorrect point classification.

1.5 Critical factor 5: Misalignment between footprints and lidar

In the preparation process of the input data, it was identified that the footprints from IGAC cartography represented the complete built area of the block, and not the individual footprints of each building. For this reason, it was necessary to carry out a subdivision process through visual interpretation using the orthophoto and parcel delimitation. During this process, it was observed that the delimitation of some buildings does not exactly coincide with the geometry captured in the lidar point cloud. As a result, mismatches occurred in the spatial correspondence between the footprints and the associated lidar points, which affected the assignment of information during the three-dimensional reconstruction process.

In one of the analyzed cases, it was observed that a building presented partial loss of volume, due to the overlap of its lidar data with the footprint of a neighboring construction. During the reconstruction process, the algorithm uses the footprint as a spatial boundary to clip the lidar point cloud; when the footprint partially invades the space of an adjacent construction, lidar points from this neighboring building are incorrectly incorporated into the modeling process. As a result, the algorithm estimates incorrect roof planes and generates inconsistent or displaced volumes, evidenced by the appearance of artificial vertical faces that connect surfaces of different buildings.

Additionally, cases of incorrectly reconstructed volume were identified; when the footprint is misaligned, lidar points belonging to a neighboring building of greater height are incorporated, causing the algorithm to reconstruct a three-dimensional volume that does not correspond to the real construction.

1.6 Critical Factor 6: Presence of vegetation

Based on the results, a building that could not be extruded had vegetation cover in its immediate surroundings. This condition generated occlusion effects during lidar data acquisition, thus reducing the effective point density over the rooftops. Consequently, the availability of information in key areas for reconstruction is limited, affecting the detection of planes and increasing the fraction of missing data. This situation directly impacts the algorithm's ability to generate an adequate reconstruction, highlighting the influence of external factors, such as vegetation, on the quality of the input data and, therefore, on the 3D model result.

In order to verify whether the lack of reconstruction was associated with the algorithm configuration and not only with data quality, additional tests were carried out adjusting different Roofer parameters. First, the *plane-detect-min-points* parameter was modified, reducing the minimum number of points required to accept a plane from 15 to 10 and then to 5; however, the building continued without being extruded. Similarly, the value of *ceil-point-density* was increased from 20 to 30, with the purpose of allowing a greater number of points in the cloud associated with the building, without obtaining favorable results. Finally, the adjustment of the *plane-detect-k* parameter was also evaluated, increasing the number of neighbors used in plane detection to favor region expansion under low-density conditions, but the building still failed to be reconstructed. Even when combining these parameters, extrusion of the building was not achieved. These results suggest that the limitation did not depend exclusively on the parameterization of the algorithm, but mainly on the absence of points on the roof, conditioned by occlusion generated by vegetation.

On the other hand, it was observed that the building that could not be extruded presented a greater number of points classified as building than other constructions that were successfully reconstructed. This suggests that the algorithm depends mainly on the spatial distribution of points over the roof, rather than on the total amount of available data. Not only the average density is determinant, but the effective density over the roof, which may be affected by external elements such as vegetation.

1.7 Critical factor 7: Estimation of base elevation in slope areas

During the analysis of the generated 3D models, inconsistencies were identified in buildings located in slope areas, associated with construction typologies common in the study area, characterized by steep slopes, such as stepped constructions, partially buried structures, or buildings with retaining walls. In this type of topographic context, the relationship between the base of the building and the surrounding terrain presents significant altimetric variations, which can affect the estimation of the base level used in three-dimensional reconstruction.

The Roofer algorithm estimates the base elevation of the building (Z_0) from points classified as ground located around the building; according to the algorithm documentation, the ground plane height is calculated using the 5th percentile of ground points located within a margin of 3 meters around the building footprint (Peters et al., 2022). This indicates that the estimation of Z_0 does not depend exclusively on ground points contained within the footprint, but also on ground points located in its vicinity.

The results reveal that, in areas with steep slope, the procedure may generate erroneous estimations of the altimetric base due to the presence of ground points with highly variable elevations within the calculation buffer. This situation is frequent in slope contexts, where one side of the building may be at road level while the other is associated with an embankment or descending slope. In addition, the inclusion of elements such as corridors, sunken patios, or ramps introduces low elevation values, affecting the estimation of the base elevation. As a result, the estimated value of Z_0 is reduced, causing the resulting 3D model to be anchored to a single elevation that does not coincide with the real base of the building.

In order to evaluate this behavior, tests were carried out by modifying the margin used for terrain estimation. Since this parameter cannot be adjusted directly in the Roofer algorithm, an external procedure was implemented by means of a script, in which the base elevation (Z_0) was previously calculated using a reduced 1 m buffer around the footprint, and subsequently, this value was incorporated as an input attribute in the reconstruction process. This adjustment made it possible to reduce the influence of ground points far from the real base of the construction, which produced an improvement in the estimation of the model base level.

Despite this improvement, the reconstructed model continues to use a single base elevation for the entire building, and therefore does not adequately represent stepped constructions or buildings adapted to slope conditions.

1.8 Critical factor 8: Polycarbonate roofs

Inconsistencies associated with buildings that present roofs constructed with translucent materials, such as polycarbonate, were identified. This type of material allows the partial transmission of the lidar signal, which generates returns both on the roof and on the terrain or surfaces located below the structure.

As a consequence, some of the returns recorded under these roofs were classified as ground points during the lidar point cloud classification process. When the Roofer algorithm uses these points to estimate the terrain base level or to adjust the roof planes, inconsistencies are produced in the reconstruction. In these cases, it was observed that the algorithm interprets the returns under the roof as open surfaces or interior patios, which generates voids or artificial cavities in the reconstructed volume.

To correct this situation, adjustments were carried out in the point cloud processing, refining the lidar classification in the areas with translucent roofs, eliminating the points that had been classified as ground under these structures. Additionally, a more automated alternative was evaluated, which consisted of modifying the clip-terrain parameter of the Roofer algorithm by setting it to false, in order to avoid the activation of the procedure that trims parts of the roof plane when patches of points classified as ground are detected; by deactivating this procedure, it is avoided those returns classified as ground under roofs generate improper clipping in the roof geometry.

The results obtained with this adjustment were equivalent to those achieved through the refinement of the lidar classification, allowing the correct reconstruction of roof geometry and eliminating the artificial voids generated in the original models. In addition, as previously mentioned, this adjustment contributes to preserving the integrity of the roof contour, generating more regular models and edges during the reconstruction process.

1.9 Critical Factor 9: Local Construction Characteristics

In the analysis of the reconstructions, limitations associated with morphological particularities of some buildings were identified, which hinder their correct representation through reconstruction methods. In particular, constructions presenting protruding architectural elements such as overhangs, balconies, parapets, and water tanks on the roofs were observed.In some cases, these protruding surfaces are artificially projected to ground level, generating non-real vertical faces. In other cases, these elements are completely omitted, producing a simplified representation of the building that does not reflect its real architectural configuration. The inaccurate representation of these buildings has direct implications in analyses derived from urban modeling, such as the calculation of built areas and volumes, where overestimations or underestimations may occur, and the evaluation of solar potential, due to alterations in slopes, orientations, and real roof surfaces.

In the context of estimating solar potential on roofs, the correct representation of constructive elements such as parapets and water storage tanks on roofs acquires special relevance. These elements occupy roof surfaces that could be used for the installation of solar panels and, in addition, generate shadows that directly influence the estimation of the available energy potential. In order to improve their representation in the 3D model, tests were carried out to reconstruct these elements using the available lidar data; however, their correct identification presented several limitations. On the one hand, many automatic lidar classification algorithms do not recognize these objects as differentiated classes, and on the other hand, the point density on these elements is insufficient, which prevents the algorithm from detecting planes that allow their adequate reconstruction.

As a result of the iterative process, which included more than 40 test configurations through adjustments to parameters and/or input data, a final model was obtained with significant improvements in geometric consistency and topological validity, reducing the inconsistencies identified in the initial stages. *Figure 1* shows the final model generated from the various iterations, demonstrating greater stability in the reconstruction and a better representation of the roofs.

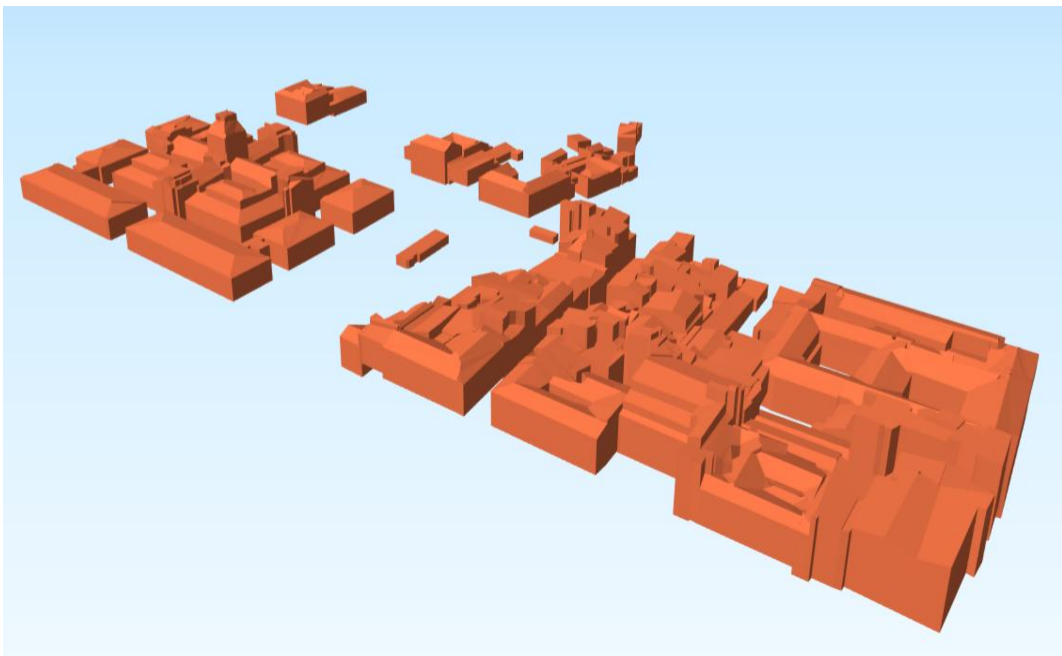

Figure 1. Final result of the 3D model after applying methodological adjustments

2. References

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